HealthBridge: A Cross-lingual Clustering Framework for English-Chinese Health News

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Abstract. Health news profoundly shapes public understanding and healthcare decisions globally, yet linguistic barriers between English and Chinese—two of the world's most widely spoken languages—create significant health knowledge disparities. To address this challenge, we propose HealthBridge, a novel cross-lingual clustering framework that automatically identifies and aligns related health articles across these languages. We present a comprehensive bilingual health news corpus comprising 68,603 articles spanning 2018–2025, with a carefully constructed evaluation set featuring 470 high-quality health news clusters. The HealthBridge framework integrates advanced text representation strategies with temporal constraints. Experimental results demonstrate that, for the task of cross-lingual health news association, methods based on complex vector representations significantly outperform explicit entity matching techniques. HealthBridge advances both cross-lingual natural language processing and global health communication by enabling systematic analysis of health reporting across linguistic boundaries. This framework not only reveals cultural variations in health news framing but also promotes equitable access to health information in an increasingly interconnected world.

Keywords: Cross-lingual Health Information · Multilingual News Clustering · Biomedical Natural Language Processing

1 Introduction

Health news fundamentally shapes health literacy across populations and drives consequential healthcare decision-making among individuals [19, 21]. As digital platforms proliferate, billions worldwide increasingly rely on health news to stay informed about medical developments, disease outbreaks, and healthcare policies. However, this expansion of health information channels has introduced significant challenges. During public health crises, inconsistent reporting standards have accelerated misinformation proliferation, undermining effective response efforts [23]. Moreover, underlying biases stemming from cultural, political, and commercial factors frequently distort health messaging, particularly affecting audiences with limited information sources [9, 40].

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Table 1. Sample health newswire articles from our dataset. Cross-lingual news coverage on red meat consumption and the risk of type 2 diabetes.

Source	Title	Publish Date
Science Net	红肉吃得越多 2 型糖尿病风险越高 (Translation: The More Red Meat Consumed, The Higher the Risk of Type 2 Diabetes)	2023–10-05
News.cn	新研究确认食用红肉与 2 型糖尿病风险的联系 (Translation: New Research Confirms Link Between Red Meat Consumption and Type 2 Diabetes Risk)	2023–10-21
NPR	Too much red meat is linked to a 50% increase in Type 2 diabetes risk	2023–10-19
Science Based Medicine	Study: Eating red meat may increase your risk of Type 2 diabetes	2023–10-23
CNN	Eating red meat linked to higher risk of type 2 diabetes, study finds	2023–10-19

Automated news clustering presents a compelling approach to address these information quality issues by systematically organizing related health articles from diverse sources [1, 4]. This methodology enables readers to develop a comprehensive understanding through exposure to multiple perspectives on the same health topic. Such comparative analysis facilitates more effective fact-checking processes and illuminates the evolution of health narratives over time. Through clustered content, readers can better distinguish credible reporting from misleading information while gaining insight into the framing techniques that shape public health perceptions [18].

While news clustering offers significant benefits for information quality assessment, linguistic barriers present a substantial challenge in our increasingly globalized health information ecosystem (see Table 1). Cross-language analysis becomes necessary as health issues extend beyond national borders, yet important information often stays trapped within single-language sources [26,11]. This language divide worsens unequal access to health knowledge and prevents a complete understanding of global health topics. Recent progress in cross-language natural language processing offers promising tools by allowing systematic comparison of health reporting across different cultures and languages [17, 24, 25]. These comparisons can show important differences in how the same health topics are presented across regions, helping to spot cultural biases while supporting fairer information access for all language backgrounds.

In this paper, we address the challenges of cross-lingual health news analysis by presenting a novel approach to multilingual news clustering. Our research contributes to this emerging field in several important ways: (1) we introduce a substantial dataset of English and Chinese health news articles that enables robust cross-lingual analysis; (2) we propose a methodological framework that combines named entity recognition, semantic representations, and temporal constraints; and (3) we demonstrate through experimental validation that our approach effectively identifies meaningful cross-lingual news clusters, facilitating comparative analysis of health reporting across language boundaries.

2 Dataset

This section presents our bilingual health news corpus comprising 68,603 articles in English and Chinese, spanning 2018–2025[1]. We constructed this dataset to support cross-lingual news clustering research specifically focused on health-domain content.

2.1 Health News Collection

English News Collection. We systematically collected 150,028 English health news articles using 15 RSS feeds from September 2018 to December 2024. Our sources included both mainstream news platforms with health sections and specialized medical news outlets, as detailed in Table 2.

For each potential article, we aimed to extract the headline, full text, and publication timestamp. However, web data acquisition often presents challenges such as broken hyperlinks or subscription barriers. Therefore, we applied specific filtering criteria. We retained an article only if its successfully extracted text exceeded 100 words and it possessed verifiable publication metadata. This filtering process resulted in a refined corpus of 48,675 English news articles.

Chinese News Collection. The Chinese component of our dataset comprises 22,055 health news articles acquired from six prominent Chinese news outlets and specialized medical information platforms (see Table 2) [44]. This corpus spans from September 2018 to February 2025, maintaining temporal alignment with our English collection. Employing consistent methodological practices, we extracted headlines, complete textual content, and publication chronology, resulting in a collection of 19,928 Chinese news articles.

During our data preprocessing phase, we identified that certain Chinese news headlines contained superfluous elements including authorial attributions and institutional identifiers. To optimize subsequent clustering operations, we implemented a systematic normalization protocol through rule-based formatting algorithms to eliminate these extraneous metadata components. To establish cross-lingual compatibility, we processed all Chinese articles through neural ma- chine translation, utilizing the Baidu Translate API[2].

2.2 Ground-Truth Evaluation Set

To evaluate the effectiveness of our cross-lingual clustering approach, we constructed a high-quality ground-truth subset of news articles that report on identical scientific research. Building on insights from Liu et al. [20] regarding news redundancy and Zuo et al. [43] on citation gaps, we observed that multiple news agencies often report on the same study, and a subset of these articles explicitly reference the original research. We developed a systematic approach to identify these naturally occurring news clusters by extracting DOI information from news articles that cited scientific studies. These unique

[1] https://github.com/chzuo/icic2025.
[2] Api.fanyi.baidu.com.

Table 2. News Sources Used for Health News Collection

English RSS Feeds	Chinese News Sources
news.google.com, reuters.com, bmj.com	cn-healthcare.com
nytimes.com, cnn.com, medpagetoday.com	news.sciencenet.cn
medicaldaily.com, medscape.com	paper.sciencenet.cn
webmd.com, news-medical.net, statnews.com	thepaper.cn
medicalnewstoday.com, news.medlive.cn	health.people.com.cn
bma.org.uk, healthcareitnews.com	

and persistent identifiers allowed us to establish definitive links between news articles reporting on identical research, regardless of language or news source.

Through our extraction and validation process, we obtained 470 high-quality health news clusters comprising 984 articles in total. Among the clusters, we identified 7 pure Chinese clusters, and 463 pure English clusters. Our initial ground-truth validation set did not contain any mixed clusters that included both Chinese and English news articles reporting on the same research, highlighting the challenge of cross-lingual news clustering. This imbalance between English and Chinese articles represents a limitation of our dataset. Additionally, the reliance on DOI-based extraction may not capture the full spectrum of health news reporting. Despite these limitations, it provides a reliable foundation for evaluating our clustering approach.

3 Experiment

This section details the experiments conducted to evaluate our proposed framework for cross-lingual health news clustering between English and Chinese. We first establish baseline models and describe the feature extraction process. Subsequently, we present the components of our framework and outline the evaluation methodology, including datasets and metrics. The results demonstrate the effectiveness of our approach in grouping news articles covering the same health events across languages compared to baseline methods.

3.1 Baseline Models and Feature Extraction

Entity-Based Similarity. The first baseline model leverages the UMLS (Unified Medical Language System) ontology to extract medical named entities from news articles [11]. For Chinese articles, we extracted entities from their translated English versions to maintain consistency in the entity space. We then computed the Jaccard similarity between news articles based on their entity sets. Considering the importance of titles in news articles, we emphasized title entities by duplicating them in the entity set, effectively increasing their weight in similarity calculations.

Semantic Vector Representation. The second approach employs Sentence-BERT [27] models to generate dense vector representations of news articles. To address the token limitation of transformer-based models and considering the inverted pyramid structure of news articles (where important information appears at the beginning), we separately encoded the title and the first 200 words of the article body, then averaged these vectors to obtain the final representation. We evaluated several model variants:

Multilingual direct processing: Using the multilingual version of MPNet [32] to process both Chinese and English articles in their original languages without translation. **English-only processing:** Using either MiniLM [34] or PubMedBERT [13] (domain-specific for biomedical content) to process English articles and translated Chinese articles.

3.2 Feature Fusion and Temporal Constraints

To leverage the complementary strengths of entity-based and semantic-based similarities, we implemented a weighted fusion approach. The final similarity between two documents d_1 and d_2 is computed as:

$$Sim(d_1, d_2) = \alpha \cdot Sim_{Jaccard}(d_1, d_2) + (1 - \alpha) \cdot Sim_{Cosine}(V_1, V_2) \tag{1}$$

where $Sim_{Jaccard}$ represents the Jaccard similarity between entity sets, Sim_{Cosine} represents the cosine similarity between document vectors V_1 and V_2, and α is the weighting parameter.

We employed DBSCAN (Density-Based Spatial Clustering of Applications with Noise) as our clustering algorithm due to its ability to discover clusters of arbitrary shapes without requiring a predefined number of clusters [31]. For implementation with DBSCAN, we convert similarities to distances using $d(d_1, d_2) = 1 - Sim(d_1, d_2)$. The algorithm operates with two key parameters: ϱ, which defines the maximum distance between two samples for them to be considered as in the same neighborhood, and x‘, which specifies the minimum number of samples in a neighborhood for a point to be considered as a core point.

Additionally, news articles reporting the same event typically appear within a limited time frame. We incorporated this temporal constraint through a time-decay penalty in the DBSCAN clustering. Instead of using a fixed distance threshold for all article pairs, we apply a time-sensitive distance adjustment:

$$d'(d_1, d_2) = d(d_1, d_2) + min(\lambda \cdot \Delta t, \lambda \cdot T_{max}) \tag{2}$$

where $d(d_1, d_2)$ is the original distance between documents d_1 and d_2, Δt is the time interval between two articles (in days), λ is a decay factor that controls how quickly the distance increases with time, and T_{max} is the maximum time difference (in days) to be considered.

All model parameters were optimized through a comprehensive grid search on our annotated ground-truth dataset. For each parameter combination, we evaluated the resulting clusters against our gold standard, selecting the configuration that maximized overall performance across the evaluation measures.

3.3 Evaluation Metrics

To evaluate clustering performance robustly against incomplete ground truth and potential noise points from DBSCAN, we employ the following metrics:

Primary Metrics: We prioritize the **Pairwise F_1 Score** (harmonic mean of Pairwise Precision (P) and Recall (R)) and the **BCubed F_1 Score** (harmonic mean of BCubed P and R). Pairwise metrics evaluate based on labeled item pairs, while BCubed metrics offer an item-centric perspective. Both are well-suited for handling incomplete annotations. We also report the respective Precision and Recall scores alongside the F1 scores for detailed analysis.
Secondary Metrics: We include the Adjusted Rand Index (ARI) as a standard reference point. However, because ARI assumes complete partitions, we calculate it only on the subset of items that are both labeled in the ground truth and assigned to a non-noise cluster by DBSCAN.

3.4 Results

The clustering evaluation results, as shown in Table 3 offer significant insights into the comparative performance of various approaches for cross-lingual health news clustering. Entity-based approaches achieve exceptional precision in BCubed metrics and ARI, indicating high cluster purity. However, this approach fundamentally suffers from an extreme pairwise precision-recall imbalance. This reveals its critical limitation: relying on exact entity matching fails to recognize semantic equivalents across languages, creating overly restrictive clusters that miss numerous valid connections. The embedding-based approaches offer more balanced performance profiles. MiniLM demonstrates the strongest individual language model, providing a superior balance between precision and recall compared to alternatives. Despite PubMedBERT's domain-specific advantages in BCubed metrics (94.92) and ARI (94.44), its lower pairwise recall (33.82) indicates limitations in capturing fine-grained cross-lingual relationships.

Table 3. Cross-lingual Health News Clustering Performance. P/R/ F1 represent Precision, Recall, and F1 Score respectively. ARI is calculated only on labeled, non-noise items.

Model	Pairwise			BCubed			ARI
	P	R	F1	P	R	F1	
UMLS Entity-based	**96.30**	3.82	7.36	98.15	84.54	90.84	**98.09**
Multilingual MPNet	67.04	43.68	52.89	94.55	93.17	93.86	78.42
MiniLM	71.86	45.44	55.68	95.40	93.44	94.41	81.50
PubMedBERT	90.55	33.82	49.25	97.71	92.30	94.92	94.44
Fusion Model	95.28	**53.38**	**68.43**	98.83	96.33	97.56	96.41
Fusion (no entities)	95.81	50.44	66.09	**99.11**	**96.46**	**97.77**	97.16

Most significantly, the Fusion Model leveraging MiniLM embeddings demonstrates substantial performance gains, confirming the effectiveness of integrating complementary approaches. Interestingly, the entity-free variant maintains comparable performance, suggesting that sophisticated vector representations can effectively bridge linguistic divides without requiring explicit entity matching. This comparison reveals a fundamental insight for cross-lingual architecture design: contextual semantic understanding that preserves conceptual integrity across linguistic boundaries can outperform rigid entity matching approaches, particularly in specialized domains where terminology varies significantly across languages.

Moreover, we analyzed the clustering results of our framework, which successfully processed a substantial corpus of 11,715 health news articles, organizing them into 4,403 distinct clusters. This represents a significant achievement in cross-lingual news clustering, with an average cluster size of 2.66 articles, though cluster sizes vary considerably (ranging from 2 to 41 articles per cluster). The language distribution reveals an asymmetry in the dataset, with English sources dominating at 8,187 articles (69.88%) compared to Chinese at 3,528 articles (30.12%). Despite this imbalance, the system successfully identified 556 mixed-language clusters (12.63% of all clusters), demonstrating effective cross- lingual bridging capabilities. The bubble chart visualization (as shown in Fig. 1) of Chinese-English ratios within mixed clusters shows intriguing patterns. Most cross-lingual clusters contain a relatively balanced small number of articles from each language (typically 2–5 articles), while a few notable outliers contain significantly more content from one language or the other. This suggests that certain health topics receive disproportionate coverage across linguistic boundaries, potentially highlighting cultural differences in health reporting priorities.

4 Related Work

Cross-lingual News Clustering has become an important research focus. Hong et al. [16] proposed a semantic correlation-based method using GVSM and Spectral Clustering to group bilingual news on the same event. Wu et al. [36] introduced a knowledge distillation framework combining sentence embeddings and topic distributions with an.

improved Single-Pass algorithm. Schneider et al. [29, 30] studied multilingual clustering at scale, revealing challenges in proper noun handling and limitations of LLMs, which often cluster by language rather than content. Miranda et al. [24] developed an efficient online approach for clustering streaming multilingual news into coherent story clusters. Belyaeva et al. [3] improved cross-lingual cluster linking using Canonical Correlation Analysis and entity vectors. Yuan and Liu [39] identified shared linguistic features in multi-lingual fake news that boost clustering performance. Building on this body of work, our study targets the health news domain and proposes a multilingual clustering framework to support comparative analysis.

Medical Text Processing is another closely related line of work. Sun et al. [33] reviewed key techniques for processing electronic medical records (EMRs), emphasizing named

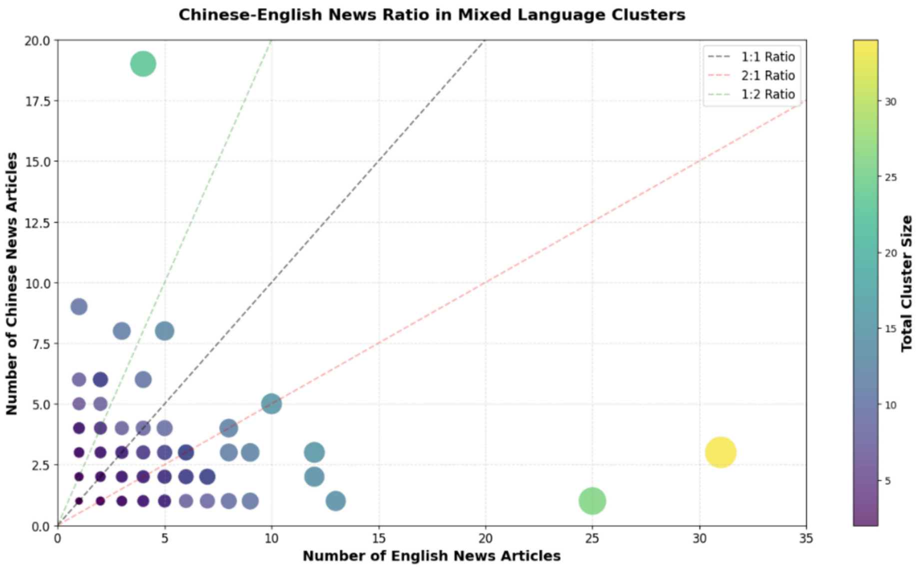

Fig. 1. Distribution of Chinese and English news articles in mixed-language clusters. Each bubble represents a distinct cluster, with bubble size and color intensity representing the total number of articles in the cluster.

entity recognition and relation extraction. Hahn and Romacker [15] showed that neglecting local coherence phenomena like anaphora can compromise knowledge representations, and proposed a prototype system to address this. Zhou and Hripcsak [41] highlighted challenges in temporal reasoning for clinical narratives, including granularity and uncertainty. Gonzalez-Hernandez et al. [12] summarized recent learning-based methods for extracting patient perspectives from EHRs and social media. Meystre and Haug [22] demonstrated improved medical problem extraction using UMLS MetaMap and negation detection. Duch et al. [10] introduced a neurolinguistic approach that enhances clustering via ontology-based concept expansion. Wilcox and Hripcsak [35] emphasized the value of expert knowledge in boosting classification performance and efficiency. Extending this line of research, our work adapts medical NLP techniques to cross-lingual health news clustering.

Cross-lingual Representation Learning is a parallel research direction. Conneau et al. [8] introduced XLM-R, a multilingual language model that outperforms mBERT across various cross-lingual benchmarks. Ruder et al. [28] surveyed weakly- and unsupervised cross-lingual word representation methods, emphasizing low-resource and distant language settings. Conneau et al. [7] and Babu et al. [2] proposed XLSR and XLS-R for multilingual speech representation, showing strong results on speech recognition and translation tasks. Xiao and Guo [37, 38] learned cross-lingual word and document representations for dependency parsing and sentiment classification. Zhou et al. [42] and Chen et al. [6] introduced models using document semantics and emojis to improve sentiment transfer across languages. Cao et al. [5] jointly modeled cross-lingual words and entities without parallel corpora, and Guo et al. [14] proposed EMMA-X, which unifies sentence representation and semantic prediction for multilingual tasks. Building on this foundation, our work applies cross-lingual representation learning to health news clustering to support multilingual health communication.

5 Conclusion

This paper introduced HealthBridge, a novel cross-lingual clustering framework designed to bridge language barriers in health news between English and Chinese. Our comprehensive dataset of 68,603 articles (2018–2025) and evaluation set of 470 news clusters enabled thorough validation of our approach. Experimental results revealed that semantic vector representations outperform entity-based methods in cross-lingual contexts, with our fusion model achieving superior clustering. The framework successfully identified 556 mixed-language clusters, enabling systematic comparison of health reporting across cultural boundaries. By illuminating framing differences while facilitating a comprehensive understanding of global health topics, HealthBridge contributes to both cross-lingual NLP research and public health communication.

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